

P1.12

The glucose-to-adipic acid process loses \$5400/day while the benzene to adipic acid process makes \$27,100. For the glucose process to be competitive, the cost for the glucose needs to drop by 27,100+5400 or by \$32,500. The current cost is \$48,500/day, so the cost would have to drop to \$16,000. At 80,850 kg/day consumption of glucose, this converts to a glucose price of \$0.198/kg.

The glucose-to-catechol process makes \$49,200/day, but the benzene-to-catechol process nets \$89,300. The difference is \$40,100. The glucose price would have to drop to \$0.104/kg to be competitive with benzene.

P1.34

This problem requires students to consider their own consumption patterns and to estimate market size based on their own consumption, and to look up information on market size in several common reference materials.

P2.11

$$1970: \left(\frac{1100 \times 10^9 \text{ mi}}{\text{year}} \right) \left(\frac{14 \text{ g}}{\text{mi}} \right) \left(\frac{1 \text{ lb}}{454 \text{ g}} \right) \left(\frac{1 \text{ ton}}{2000 \text{ lb}} \right) = 17 \text{ million tons/yr}$$

$$2000: \left(\frac{2600 \times 10^9 \text{ mi}}{\text{year}} \right) \left(\frac{1 \text{ g}}{\text{mi}} \right) \left(\frac{1 \text{ lb}}{454 \text{ g}} \right) \left(\frac{1 \text{ ton}}{2000 \text{ lb}} \right) = 2.9 \text{ million tons/yr}$$

$$(17-2.9)/17 \times 100\% = 83\% \text{ decrease in emissions.}$$